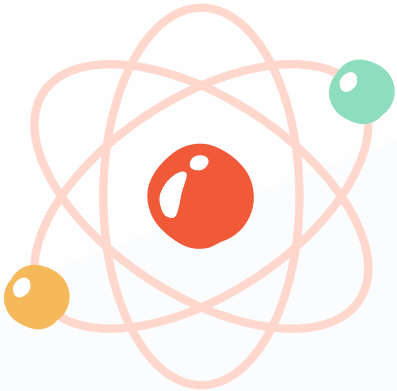
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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Modern Physics-2

English Medium

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MODERN PHYSICS-2

KEY CONCEPT

NUCLEAR PHYSICS

ATOMIC NUCLEUS

The atomic nucleus consists of two types of elementary particles, viz. protons and neutrons. These particles are called nucleons. The proton (denoted by p) has a charge +e and a mass $m_p = 1.6726 \times 10^{-27}$ kg, which is approximately 1840 times larger than the electron mass. The proton is the nucleus of the simplest atom with Z = 1, viz the hydrogen atom.

The neutron (denoted by n) is an electrically neutral particle (its charge is zero). The neutron mass is 1.6749×10^{-27} kg. The fact that the mass of a neutron exceeds the mass of a proton by about 2.5 times the electronic masses is of essential importance. It follows from this that the neutron in free state (outside the nucleus) is unstable (radioactive). With half life equal to 12 min, the neutron spontaneously transforms into a proton by emitting an electron (e^-) and a particle called the antineutrino $(\bar{\nu})$.

This process can be schematically written as follows : ${}_0n^1 \rightarrow {}_1p^1 + {}_{-1}e^0 + \bar{\nu}$

The most important characteristics of the nucleus are the charge number Z (coinciding with atomic number of the element) and mass number A. The charge number Z is equal to the number of protons in the nucleus, and hence it determines the nuclear charge equal to Ze. The mass number A is equal to the number of nucleons in the nucleus (i.e., to the total number of protons and neutrons). Nuclei are symbolically designated as $X_Z{}^A$ or ${}_ZX^A$ where X stands for the symbol of a chemical element.

For example, the nucleus of the oxygen atom is symbolically written as O_8^{16} or ${}_8O^{16}$.

The shape of nucleus is approximately spherical and its radius is approximately related to the mass number by

$R = 1.2\ A^{1/3} \times 10^{-15}\ m = 1.2 \times 10^{-15} \times A^{1/3}\ m$

Most of the chemical elements have several types of atoms differing in the number of neutrons in their nuclei. These varieties are called isotopes. For example carbon has three isotopes ${}_6C^{12}$, ${}_6C^{13}$, ${}_6C^{14}$. In addition to stable isotopes, there also exist unstable (radioactive) isotopes. Atomic masses are specified in terms of the atomic mass unit or unified mass unit (u). The mass of a neutral atom of the carbon ${}_6C^{12}$ is defined to be exactly 12 u.

$1u = 1.66056 \times 10^{-27}$ kg = 931.5 MeV.

BINDING ENERGY

The rest mass of the nucleus is smaller than the sum of the rest masses of nucleons constituting it. This is due to the fact that when nucleons combine to form a nucleus, some energy (binding energy) is liberated. The binding energy is equal to the work that must be done to split the nucleus into the particles constituting it.

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The difference between the total mass of the nucleons and mass of the nucleus is called the mass defect of the nucleus represented by $\Delta m = [Zm_p + (A - Z)m_n] - m_{nuc}$

Multiplying the mass defect by the square of the velocity of light, we can find the binding energy of the nucleus.

$$BE = \Delta mc^2 = [(Zm_p + (A - Z)m_n) - m_{nuc}]c^2$$

If the masses are taken in atomic mass unit, the binding energy is given by

$$BE = [(Zm_p + (A - Z)m_n) - m_{nuc}]\ 931.5\ MeV$$

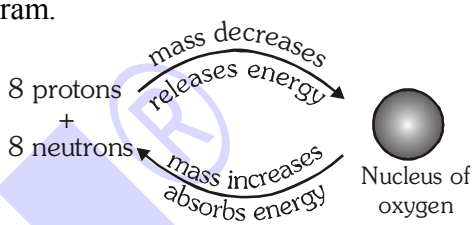
Let us take example of oxygen nucleus. It contains 8 protons and 8 neutrons.

We can discuss concept of binding energy by following diagram.

$8m_p + 8m_n >$ mass of nucleus of oxygen

For nucleus we apply mass energy conservation,

$$8m_p + 8m_n = \text{mass of nucleus} + \frac{B.E.}{c^2}$$



For general nucleus ${}^{A}_{Z}X$, mass defect = difference between total mass of nucleons and mass of the nucleus

$$\Delta m = [Zm_p + (A - Z)m_n] - M$$

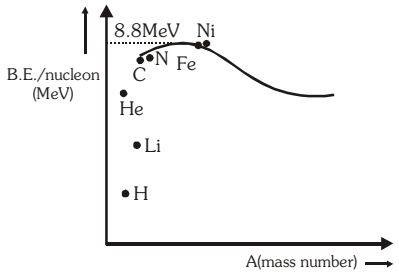
$$B.E. = \Delta mc^2 \text{ (joules)} = (\Delta m)_{\text{in amu}} \times 931.5\ MeV$$

Binding Energy per Nucleon

Stability of a nucleus does not depend upon binding energy of a nucleus but it depends upon binding energy per nucleon

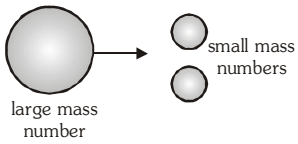
$$B.E./\text{nucleon} = \frac{B.E.}{\text{mass number}}$$

$$\text{Stability} \propto \frac{B.E.}{A}$$

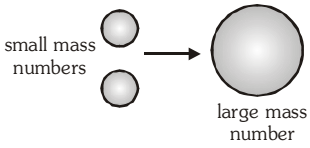


(i) B.E./A is maximum for A =62 (Ni), It is 8.79460 ± 0.00003 MeV/nucleon, means most stable nuclei are in the region of A = 62.

(ii) Heavy nuclei achieve stability by breaking into two smaller nuclei and this reaction is called fission reaction.



(iii) Nuclei achieve stability by combining and resulting into heavy nucleus and this reaction is called fusion reaction.



(iv) In both reactions products are more stable in comparison to reactants and Q value is positive.

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NUCLEAR COLLISIONS

We can represent a nuclear collision or reaction by the following notation, which means X (a,b) Y

$$\underset{\text{(bombarding particle)}}{a} + \underset{\text{(at rest)}}{X} \rightarrow Y + b$$

We can apply :

(i) Conservation of momentum (ii) Conservation of charge (iii) Conservation of mass–energy

For any nuclear reaction $\underset{K_1}{a} + \underset{K_2}{X} \rightarrow \underset{K_3}{Y} + \underset{K_4}{b}$

By mass energy conservation

(i) $K_1 + K_2 + (m_\alpha + m_x)c^2 = K_3 + K_4 + (m_Y + m_b)c^2$

(ii) Energy released in any nuclear reaction or collision is called Q value of the reaction

(iii) $Q = (K_3 + K_4) - (K_1 + K_2) = \Sigma K_P - \Sigma K_R = (\Sigma m_R - \Sigma m_P)c^2$

(iv) If Q is positive, energy is released and products are more stable in comparison to reactants.

(v) If Q is negative, energy is absorbed and products are less stable in comparison to reactants.

$Q = \Sigma(B.E.)_{product} - \Sigma(B.E.)_{reactants}$

NUCLEAR FISSION

In 1938 by Hahn and Strassmann. By attack of a particle splitting of a heavy nucleus (A > 230) into two or more lighter nuclei. In this process certain mass disappears which is obtained in the form of energy (enormous amount)

$$A + p \rightarrow \text{excited nucleus} \rightarrow B + C + Q$$

Hahn and Strassmann done the first fision of nucleus of U^{235}).

When U^{235} is bombarded by a neutron it splits into two fragments and 2 or 3 secondary neutrons and releases about 190 MeV ($\simeq$ 200 MeV) energy per fission (or from single nucleus)

Fragments are uncertain but each time energy released is almost same.

Possible reactions are

$U^{235} + {}_0n^1 \rightarrow Ba + Kr + 3\,{}_0n^1 + 200$ MeV or $U^{235} + {}_0n^1 \rightarrow Xe + Sr + 2\,{}_0n^1 + 200$ MeV

and many other reactions are possible.

- The average number of secondary neutrons is 2.5.
- Nuclear fission can be explained by using "liquid drop model" also.
- The mass defect Δm is about 0.1% of mass of fissioned nucleus
- About 93% of released energy (Q) is appear in the form of kinetic energies of products and about 7% part in the form of γ – rays.

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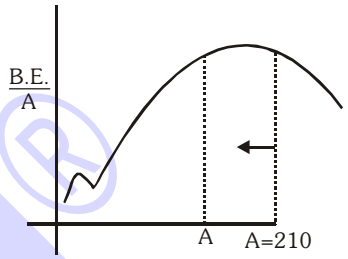
RADIOACTIVITY

The process of spontaneous disintegration shown by some unstable atomic nuclei is known as natural radioactivity. This property is associated with the emission of certain types of penetrating radiations, called radioactive rays, or Becquerel rays (α, β, γ–rays). The elements or compounds, whose atoms disintegrate and emit radiations are called radioactive elements. Radioactivity is a continuous, irreversible nuclear phenomenon.

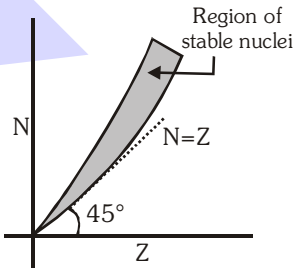
Radioactive Decays

Generally, there are three types of radioactive decays (i) α decay (ii) β^- and β^+ decay (iii) γ decay

- **α decay :** In α decay, the unstable nucleus emits an α particle. By emitting α particle, the nucleus decreases it's mass energy number and move towards stability. Nucleus having A>210 shows α decay.By releasing α particle, it can attain higher stability and Q value is positive.



- **β decay :** In beta decay (N/Z) ratio of nucleus is changed. This decay is shown by unstable nuclei. In beta decay, either a neutron is converted into proton or proton is converted into neutron. For better understanding we discuss N/Z graph. There are two type of unstable nuclides



  - **A type**

    For A type nuclides $(N/Z)_A > (N/Z)$ stable

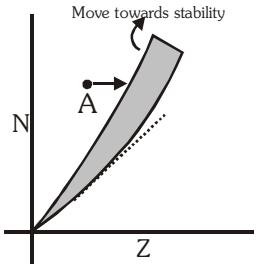
    To achieve stability, it increases Z by conversion of neutron into proton

    $${}_0n^1 \rightarrow {}_1p^1 + e^{-1} + \bar{\nu},\ {}_ZX^A \rightarrow {}_{Z+1}Y^A + \underset{(\beta\ \text{particle})}{e^{-1}} + \bar{\nu}$$

    This decay is called β^{-1} decay.

    Kinetic energy available for e^{-1} and $\bar{\nu}$ is, $Q = K_\beta + K_{\bar{\nu}}$

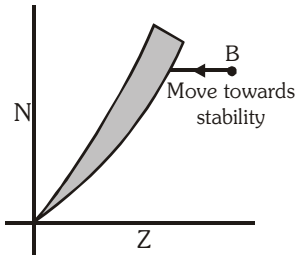
    K.E. of β satisfies the condition $0 < K_\beta < Q$



  - **B type**

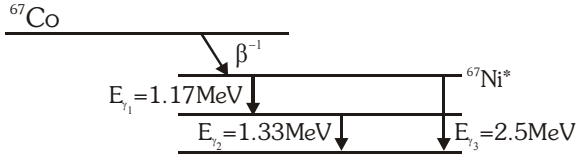
    For B type nuclides $(N/Z)_B < (N/Z)$ stable

    To achieve stability it decreases Z by the conversion of a proton into neutron. That is, $p \rightarrow n + \underset{(\text{positron})}{e^+} + \underset{(\text{neutrino})}{\nu},\ {}_ZX^A \rightarrow {}_{Z-1}Y^A + \underset{(\beta^+\text{particle})}{e^+} + \nu$



- **γ decay :** When an α or β decay takes place, the daughter nucleus is usually in higher energy state, such a nucleus comes to ground state by emitting a photon or photons.

  Order of energy of γ photon is 100 KeV



  e.g. $^{67}_{27}Co \rightarrow \underset{(\text{higher energy state})}{^{67}_{28}Ni^*} + \beta^- + \bar{\nu},\ {}^{67}_{28}Ni^* \rightarrow {}^{67}_{28}Ni + \gamma$ photon

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Properties of α, β and γ rays

Features	**α–particles**	**β–particles**	**γ–rays**
Identity	Helium nucleus or doubly ionised helium atom ($_2He^4$)	Fast moving electrons ($_{-}\beta^0$ or β^-)	Electromagnetic wave (photons)
Charge Mass	Twice of proton (+2e) $\approx 4m_p$ m_p–mass of proton	Electronic (– e) (rest mass of β) = (rest mass of electron)	Neutral rest mass = 0
Speed	1.4×10^7 m/s. to 2.2×10^7 m/s. (Only certain value between this range). Their speed depends on nature of the nucleus. So that it is a characteristic speed.	1% of c to 99% of c (All possible values between this range) β–particles come out with different speeds from the same type of nucleus. So that it can not be a characteristic speed.	Only $c = 3 \times 10^8$ m/s γ–photons come out with same speed from all types of nucleus. So, can not be a characteristic speed.
K.E.	≈ MeV	≈ MeV	≈ MeV
Energy spectrum	Line and discrete (or linear)	Continuous (or linear)	Line and discrete
Ionization power (α>β>γ)	10,000 times of γ–rays	100 times of γ–rays (or $\frac{1}{100}$ times of α)	1 (or $\frac{1}{100}$ times of β)
Penetration power (γ>β>α)	$\frac{1}{10000}$ times of γ–rays	$\frac{1}{100}$ times of γ–rays (100 times of α)	1(100 times of β)
Effect of electric or magnetic field	Deflection	Deflection (More than α)	No deflection
Explanation of emission	By Tunnel effect (or quantum mechanics)	By weak nuclear interactions	With the help of energy levels in nucleus

Laws of Radioactive Decay

1. The radioactive decay is a spontaneous process with the emission of α, β and γ rays. It is not influenced by external conditions such as temperature, pressure, electric and magnetic fields.

2. The rate of disintegration is directly proportional to the number of radioactive atoms present at that time i.e., rate of decay $\propto$ number of nuclei.

Rate of decay = λ (number of nuclei) i.e. $\frac{dN}{dt} = -\lambda N$

where λ is called the decay constant. This equation may be expressed in the form $\frac{dN}{N} = -\lambda dt$.

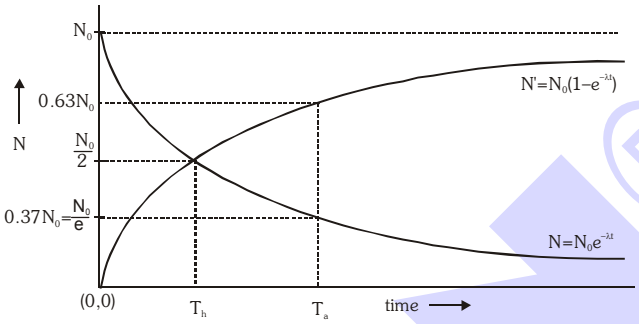
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$$\int_{N_0}^{N}\frac{dN}{N}=-\lambda\int_0^t dt \Rightarrow \ell n\left(\frac{N}{N_0}\right)=-\lambda t$$

where N_0 is the number of parent nuclei at t = 0. The number that survives at time t is therefore

$N = N_0e^{-\lambda t}$ and $t=\frac{2.303}{\lambda}\log_{10}\left(\frac{N_0}{N_t}\right)$ this function is plotted in figure.

Graph : Time versus N (or N')

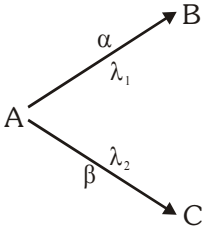


- **Parallel radioactive disintegration**

Let initial number of nuclei of A is N_0 then at any time number of nuclei of

A, B & C are given by $N_0 = N_A + N_B + N_C \Rightarrow \frac{dN_A}{dt}=\frac{-d}{dt}(N_B+N_C)$

A disintegrates into B and C by emitting α, β particle.



Now, $\frac{dN_B}{dt}=-\lambda_1N_A$ and $\frac{dN_C}{dt}=-\lambda_2N_A \Rightarrow \frac{d}{dt}(N_B+N_C)=-(\lambda_1+\lambda_2)N_A$

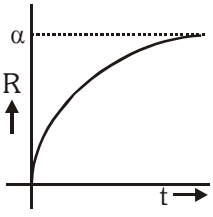
$$\Rightarrow \frac{dN_A}{dt}=-(\lambda_1+\lambda_2)N_A \Rightarrow \lambda_{eff}=\lambda_1+\lambda_2 \Rightarrow t_{eff}=\frac{t_1t_2}{t_1+t_2}$$

Radioactive Disintegration with Successive Production

$$\xrightarrow[(\alpha=\text{rate of production})]{\alpha} A \xrightarrow{\lambda} B$$

$$\frac{dN_A}{dt}=\alpha=\lambda N_A \ldots.(i)$$

when N_A in maximum $\frac{dN_A}{dt}=0=\alpha-\lambda N_A=0$, $N_A \max=\frac{\alpha}{\lambda}=\frac{\text{rate of production}}{\lambda}$



By equation (i) $\int_0^t\frac{dN_A}{\alpha-\lambda N_A}=\int_0^t dt$, Number of nuclei is $N_A=\frac{\alpha}{\lambda}(1-e^{-\lambda t})$

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